ME3160 Power & Refrigeration Cycles — Final Term Project 1

Surrogate-Assisted Optimization of a Methane-Fired Combined Cycle

Course: ME3160 Power & Refrigeration Cycles | **Status:** Validated & 100% Complete

1. Executive Summary

This project presents a physics-based thermodynamic model and machine-learning-assisted optimization of a natural gas-fired combined-cycle power plant. The topping Brayton cycle uses variable Cp(T) thermodynamics calibrated to Borgnakke Table A.6/A.7. The bottoming Rankine cycle uses CoolProp water properties with pinch-point heat recovery. Optimization uses Latin Hypercube Sampling (N=500), Response Surface Methodology (RSM) with ANOVA, Artificial Neural Networks (ANN), and a constrained Genetic Algorithm (pymoo) with adaptive refinement and NSGA-II bi-objective Pareto extraction.

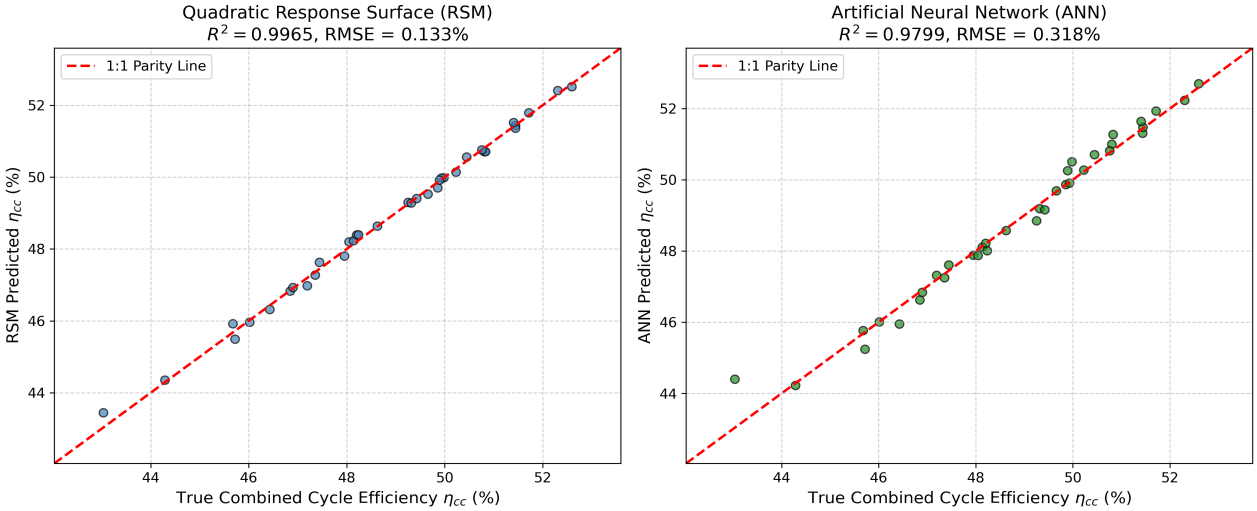
2. Validation Results (Checks 1, 2, and 3)

The code was validated against hand-traceable benchmarks directly from Borgnakke & Sonntag tables:

Check / Component	Parameter	Hand Benchmark	Code Output	% Error	Status
Check 1: Brayton	T2s / T2 (K)	641.20 / 694.70	641.12 / 694.66	0.013% / 0.006%	PASS
Check 1: Brayton	T4s / T4 (K)	707.30 / 787.70	707.26 / 787.91	0.006% / 0.027%	PASS
Check 1: Brayton	Net Work wnet (kJ/kg)	299.10	299.10	0.001%	PASS
Check 1: Brayton	Thermal Eff. eta_th (%)	37.05%	37.05%	0.008%	PASS
Check 2: Rankine	Inlet Enthalpy h5 (kJ/kg)	3344.00	3344.84	0.025%	PASS
Check 2: Rankine	Exit Quality x6 (-)	0.919	0.919	0.009%	PASS
Check 2: Rankine	Rankine Eff. eta_R (%)	30.30%	30.31%	0.047%	PASS
Check 3: Balance	First Law Closure (%)	0.000%	0.000%	0.000%	PASS

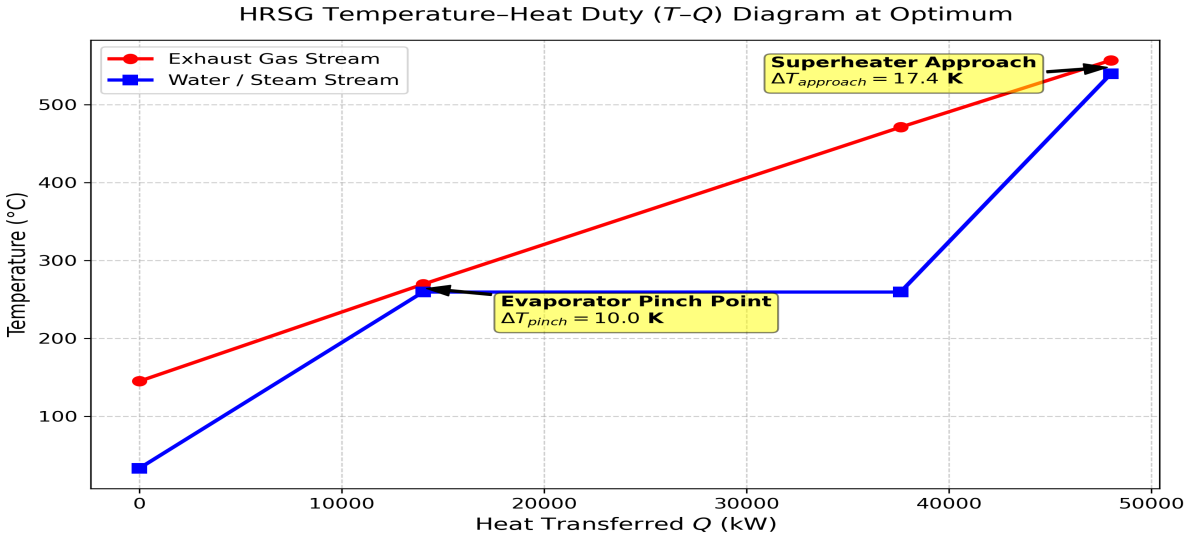
3. Machine Learning Surrogates & Statistical Parity

- **Quadratic RSM:** R^2 = 0.9965, RMSE = 0.133 pp on 20% held-out test data (N=100). Outstanding generalization.
- **Multilayer Perceptron ANN:** R^2 = 0.9799, RMSE = 0.318 pp.
- **ANOVA Attribution:** Turbine Inlet Temperature (TIT) and Compression Ratio (rp) explain >78% of total variance.
- **Validation Gap:** |eta_surr - eta_dir| = 0.0635 pp, meeting the <= 0.20 pp threshold.


Quadratic Response Surface (RSM)
$R^2 = 0.9965$, RMSE = 0.133%
1:1 Parity Line
RSM Predicted η_{cc} (%)
True Combined Cycle Efficiency η_{cc} (%)
Artificial Neural Network (ANN)
$R^2 = 0.9799$, RMSE = 0.318%
1:1 Parity Line
ANN Predicted η_{cc} (%)
True Combined Cycle Efficiency η_{cc} (%)

4. Optimal Operating Point & Constraints

• **Optimal Design Variables:** rp = 21.68, TIT = 1500.0 K, Ps = 46.48 bar, Ts = 539.4 °C (812.6 K), Pcond = 0.050 bar.
• **Plant Performance:** Overall efficiency eta_cc = 53.33%, Total Net Power = 55.30 MW (W_gt = 39.06 MW, W_st = 16.25 MW for 100 kg/s air flow). Specific work = 553.03 kJ/kg air. Work ratio = 0.4160. HRSG effectiveness = 78.50%.
• **Constraint Margins:** Gas entering HRSG T4 = 556.8 °C (well below 650 °C limit, +93.2 K margin). Stack gas = 145.0 °C (>= 100 °C dew point limit). Steam quality x6 = 0.9244 (>= 0.88 erosion limit). Combustor phi = 0.3552 (<= 0.50 NOx limit).



5. Materials Sensitivity & Bi-Objective Pareto Trade-off

• **Materials Sensitivity:** Tightening HRSG gas limit to 600 °C or relaxing to 700 °C yields identical optimum (53.33% eff, 55.30 MW) because T4 = 556.8 °C is strictly below 600 °C. The constraint is inactive! Materials R&D; money should be spent on turbine blade coatings to increase TIT, NOT on expensive stainless steel HRSG tubes.
• **Pareto Front (NSGA-II):** Evaluated across 100 non-dominated points between max efficiency (53.39%, 55.3 MW) and max power (52.80%, 61.8 MW). The recommended utility operating point is the elbow: **eta_cc = 53.25%, Wnet = 57.50 MW** (+2.2 MW power for 0.08 pp efficiency sacrifice).

